

1. Right triangle PYZ has vertices at $P(9,7)$, $Y(3,-1)$, $Z(9,-1)$. Right triangle RST has vertices at $R(6,-2)$, $S(1,4)$, $T(1,-2)$.
 - a. Determine if $\triangle PYZ$ and $\triangle RST$ are congruent, similar, or neither.
 - b. What definition, theorem, or property justifies your conclusions?
 - c. Describe the transformations that would show that the figures are congruent or that they are similar. State which image you used as the pre-image.
2. Determine if the given polygons are congruent, similar, or neither.
 - a. Trapezoid $FDGK$: $F(-8,-2)$, $D(-4,-2)$, $G(-2,-8)$, $K(-10,-6)$.
Trapezoid $MPRT$: $M(8,2)$, $P(4,2)$, $R(2,8)$, $K(10,6)$
 - ☐ Congruent
 - ☐ Similar
 - ☐ Neither

- b. Kite $ABCD$: $A(-1,1)$, $B(4,1)$, $C(5,4)$, $D(2,5)$.
Kite $KFRP$: $K(0,0)$, $F(-1,-4)$, $R(-3,-3)$, $P(-4,-1)$.

- ☐ Congruent
- ☐ Similar
- ☐ Neither

- c. Rectangle $JKMP$: $J(-4,4)$, $K(0,-2)$, $M(3,0)$, $P(-1,6)$.
Rectangle $BCDF$: $B(8,6)$, $C(0,8)$, $D(-1,4)$, $F(7,2)$

- ☐ Congruent
- ☐ Similar
- ☐ Neither

3. Hannah and Zuki are determining if parallelograms $NPSE$ and $FGHR$ are congruent, similar, or neither. Parallelogram has vertices at $N(-9,2)$, $P(-1,5)$, $S(1,9)$, and $E(-7,6)$. Parallelogram $F(-4.5,1)$, $G(-0.5,2.5)$, $H(0.5,4.5)$, and $R(-3.5,3)$. Their work is shown.

Hannah	Zuki
Parallelograms $NPSE$ and $FGHR$ are similar because	Parallelograms $NPSE$ and $FGHR$ are not similar because
$\frac{NE}{FR} = \frac{PS}{GH}$	$\frac{NP}{FG} = \frac{SE}{HR}$

- a. Who do you agree with?
- b. Explain the mistake that was made and how to correct it.